predominantly public networks and not private networks with an enterprise. Similarly, cloud is also not physically present within an enterprise.

You cannot be certain of the data centre location. Even if you did know about its location, you will still not be able to know the amount of latency it will take, because it is going through the web network and not through the public network. This is why this latency needs to be changed.

With the introduction of 5G this can be dealt with easily. It allows you to create an alternative Wi-Fi network within the enterprises. It ensures low latency and high throughput as well as privacy.

Along with that, you also need to have a miniature version of the cloud, which we call the 'H' cloud. It is more reliable for things like augmented reality (AR) and virtual reality (VR), video analytics, robotic operations, and autonomous vehicles, as it guarantees low latency. It is also important to note that the cloud is essentially for bookkeeping and data management and not for any computation tasks.

Information technology (IT) applications such as Zoom, system applications and products (SAP) can be run properly on Wi-Fi but operational applications cannot. This is almost a 517 billion dollar market, which will be disrupted eventually.

Industry 4.0 use cases

Industry 4.0 has many use cases, some of which are mentioned below.

Logistics: Warehouses tend to be extremely huge, hence requiring the need for digitising them as real-time tracking of the devices across the warehouse is extremely important. Drones can be used for tracking purposes while robotic automations can assist with moving the goods from one place to another. AR/VR based applications can be used to assist the training of the employees in real time.

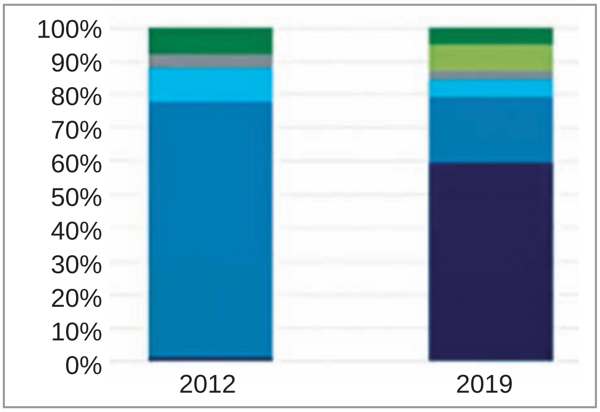

Figure 1: Data centre networking transformation

Mining: Mining in India is of two kinds. First one is called open-pit mining. This kind of mining does not require you to get into a tunnel. You essentially do the mining in the periphery and do the extraction.

The other kind of mining is called closed-pit mining. This method of mining requires you to go underground inside a tunnel. There can be around 10 to 22 tunnels as you go down, and each of them can be one to two kilometres long.

In places like Australia, a lot of automation has been done already. But in India things are still done manually. Miners enter closed pits and use a USB stick to take the data out from the sensors installed in that area. Only then can they connect it to their office systems and servers to use it.

This creates opportunities to digitise everything. By automating the unmanned vehicles this problem of data extraction can be solved so that no human has to enter these mines. This would significantly reduce the risk of any kind of disasters.

You cannot send people inside the mines for causing a blast for a specific purpose. In such cases you can send drones equipped with 4K cameras and take a survey of that area.

Oil and gas: Oil and gas exploration is divided into three predominant parts, namely, upstream - offshore, midstream - pipeline, and downstream - refinery. During upstream - offshore exploration you are basically looking for places where you can find extra oil. Similarly, there is onshore exploration where the exploration is done on land, unlike offshore.

Oil and gas exploration requires a lot of systematic data collection. Heavy analytics is required to find out the right places to conduct these explorations. Private 5G can play an important role to achieve the same. This makes a huge scope to increase productivity and save lives.

Manufacturing: There is a real use case at Fujitsu where they use private 5G network as well as edge computing for autonomous guided vehicles. They use 4K cameras to ensure that these vehicles are moving properly. They collect the data from these 4K cameras as well as sensors, which allows them to know about their real-time movement.

Similarly, in the manufacturing assembly lines they use cameras to ensure the work is being done the right way. They use artificial intelligence (AI) analytics tools to monitor everything. If they find that someone is not doing the work correctly, the AI gives a proper feedback about it. Training is also being done on the go with the help of AI in real time.

Transformation towards openness and disaggregation

Data centres of companies like Facebook and Google have moved to enable openness and disaggregation, especially in the networking space. Back in 2012, networking within most of the data centres that required switches, routers, and other network appliances were all properties, which